

Dynamics of Mitochondrial Transfer in Tumors

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Abstract

Mitochondrial transfer within the tumor microenvironment (TME) represents a key indicator of immune evasion and cancer progression. Recent studies have demonstrated that cancer cells can acquire functional mitochondria from immune cells to restore oxidative phosphorylation (OXPHOS) and promote survival. In contrast, other reports describe transfer in the opposite direction, where tumor-derived mutated mitochondria are delivered to immune cells, making immune cells dysfunctional. This highlights a complex mode of mitochondrial transfer within the TME.

In our analysis, we aim to infer mitochondrial transfer by examining mixed mitochondrial variant signatures. We replicated the findings of Zhang *et al.* (1) using their computational tool MERCI to investigate mitochondrial exchange between tumor cells and T cells. Furthermore, we have identified suitable datasets to further explore mitochondrial transfer using MERCI (2). These findings will contribute to a growing understanding of how intercellular crosstalk shapes tumor-immune interactions and may reveal new vulnerabilities for therapeutic targeting and prognosis.

Introduction

Mitochondria are best known for orchestrating oxidative phosphorylation, yet their functional repertoire extends well beyond ATP production to include regulation of apoptosis, reactive oxygen species balance, and innate immune signaling. Over the past decade it has become evident that mitochondria are not confined to the cytoplasm of their cell of origin. Instead, organelles or mitochondrial DNA fragments can be horizontally transferred between neighboring cells through tunneling nanotubes, gap junction-like structures, cell fusion events, and extracellular vesicles loaded with mitochondrial cargo.

Solid tumors are densely populated ecosystems in which malignant, stromal, and immune cell populations compete for oxygen and nutrients. Within this milieu mitochondrial transfer has been implicated as a metabolic barter system: stressed cancer cells lacking intact oxidative phosphorylation can acquire healthy mitochondria from infiltrating immune cells, while damaged or mutated mitochondrial genomes may traffic in the opposite direction, thereby attenuating immune surveillance. Understanding the prevalence, directionality, and functional impact of such transfers across tumor types and stages is essential because they may explain heterogeneity in therapeutic response to immune checkpoint blockade,

chimeric antigen receptor T-cell therapy, and metabolic inhibitors.

First, we seek to replicate and extend the MERCI benchmark analysis to validate our computational pipeline. Second, we quantify mitochondrial coverage and variant burden to establish detection sensitivity across single-cell RNA sequencing platforms. Third, we interrogate donor-receiver dynamics in coculture systems representative of early versus advanced disease. Finally, we outline a roadmap for scaling these analyses to publicly available datasets spanning multiple cancer types and integrating single-cell multi-omics.

Results

Replication of MERCI Benchmarking Analysis

We successfully replicated the benchmark analysis from the MERCI paper using their publicly available code and dataset (1). The analysis focused on the scRNA-seq Bench1 dataset, which contains cocultured (CC) with T cells, and monocultured (MC) cancer cells.

First, we analyzed the coverage of the mitochondrial genome in the sample. Figure 1 illustrates the read coverage distribution across different sequencing depths (1x to 100x) for the mitochondrial genome. This coverage analysis is critical for quality control in single-cell mitochondrial studies, as sufficient coverage is necessary to reliably detect mt-SNVs. The decreasing trend indicates the inherent challenge of achieving deep, uniform mitochondrial genome coverage in standard 10x Genomics scRNA-seq data, which motivated the development of specialized protocols like mtscATAC-seq for improved mitochondrial DNA profiling in the paper.

Figure 2 shows the distribution of mitochondrial weight metrics in the different cells. The MT-Donor-index represents the relative abundance of donor (T cell)-derived mitochondrial DNA variants in each cell. The MT-Receiver-index indicates the likelihood score of a cell being a mitochondrial receiver based on the presence of T cell-enriched mtSNVs. MT-Receiver-frac shows higher values in cocultured cancer cells compared to monocultured cells, confirming that cancer cells acquire T cell mitochondria during coculture.

Methods

MERCI (Mitochondrial-Enabled Reconstruction of Cellular Interactions) method is a computational tool that traces mitochondrial transfer between T cells and cancer cells using single-cell RNA sequencing data. The framework consists

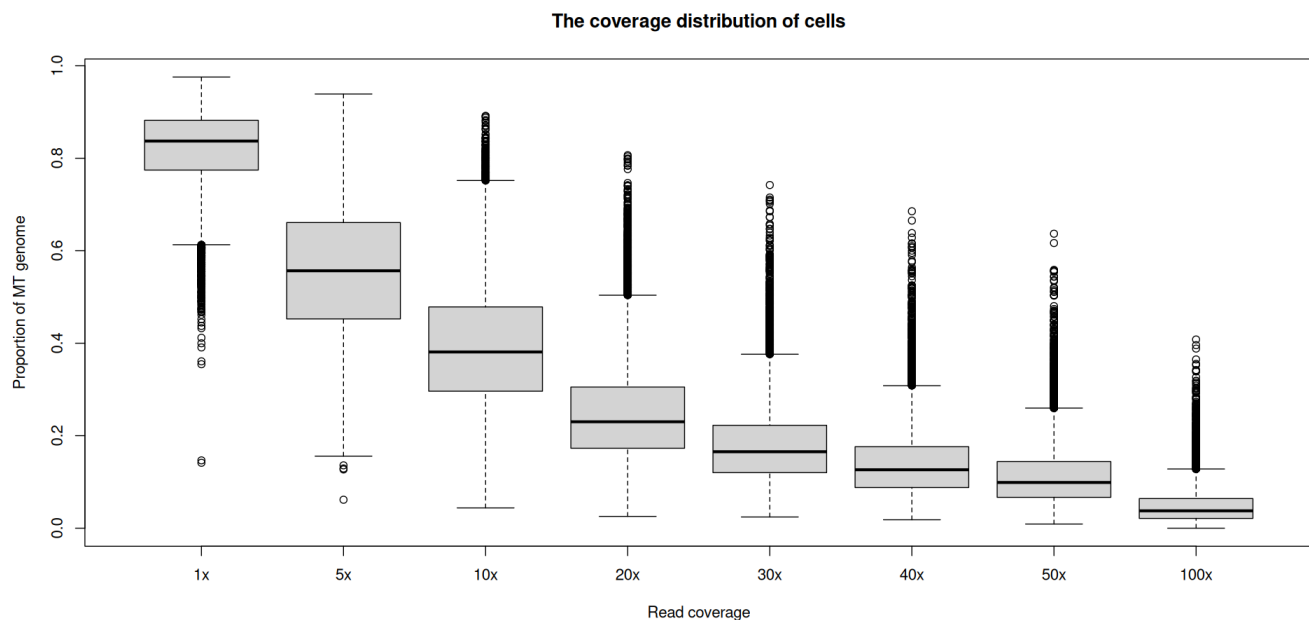


Figure 1. Boxplots showing the distribution of read coverage per cell scRNA-seq dataset (Bench1). Box center line: median; box limits: upper and lower quantiles;

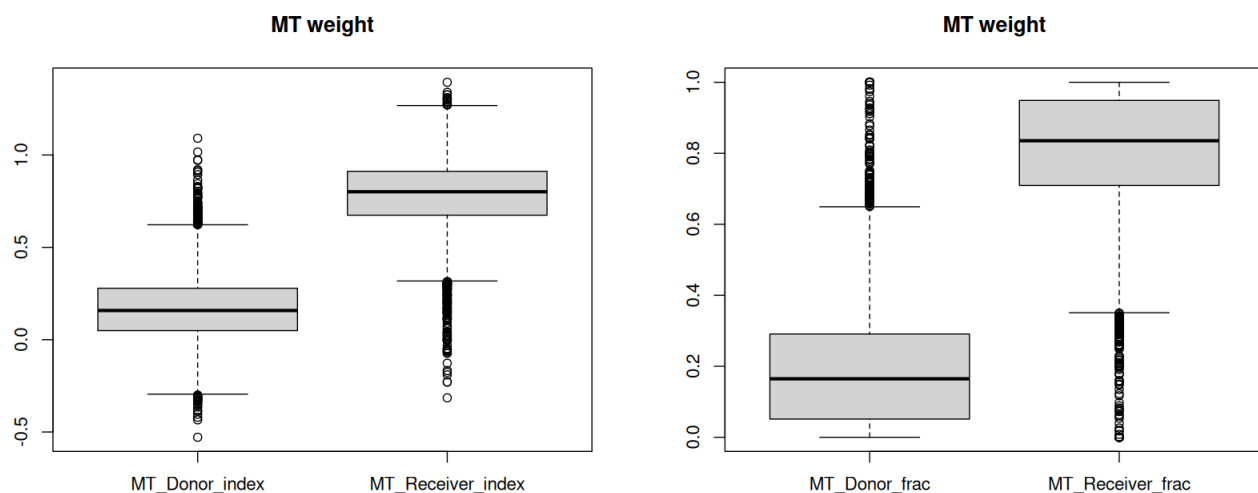


Figure 2. Plots distribution of mitochondrial weight metrics across cells (Bench1). Box center line: median; box limits: upper and lower quantiles; box whiskers: 1.5× interquartile range (IQR).

of two main modules: MERCI-mtSNP for calling mitochondrial single nucleotide variants (mtSNVs) from sequencing data (BAM files are needed in particular), and the MERCI R package for predicting mitochondrial receiver cells and estimating their mitochondrial composition.

Their data set contains single cell RNA sequencing data from three experimental groups: cocultured cancer cells and CD8+ T cells serving as donors (CC group), monocultured cancer cells without exposure to T cells (MC group). The co-culture system allows T cells to transfer mitochondria to cancer cells through mechanisms such as tunneling nanotubes, providing an ideal experimental setup to study mitochondrial transfer dynamics.

MERCI-mtSNP is a Python module that identifies mt-

SNVs that are enriched in specific cell populations (e.g., T cells versus cancer cells), which serve as genetic barcodes to trace mitochondrial origins. Following variant calling, we use the MERCI R package to predict which cells are mitochondrial receivers and estimate their mitochondrial composition. The pipeline employs two complementary approaches: DNA-based detection (DNA rank) and RNA-based detection (RNA rank). We replicated the mitochondrial weight distribution analysis and coverage assessment using the provided code and visualization scripts from the original publication.

MERCI Rough Workflow Overview

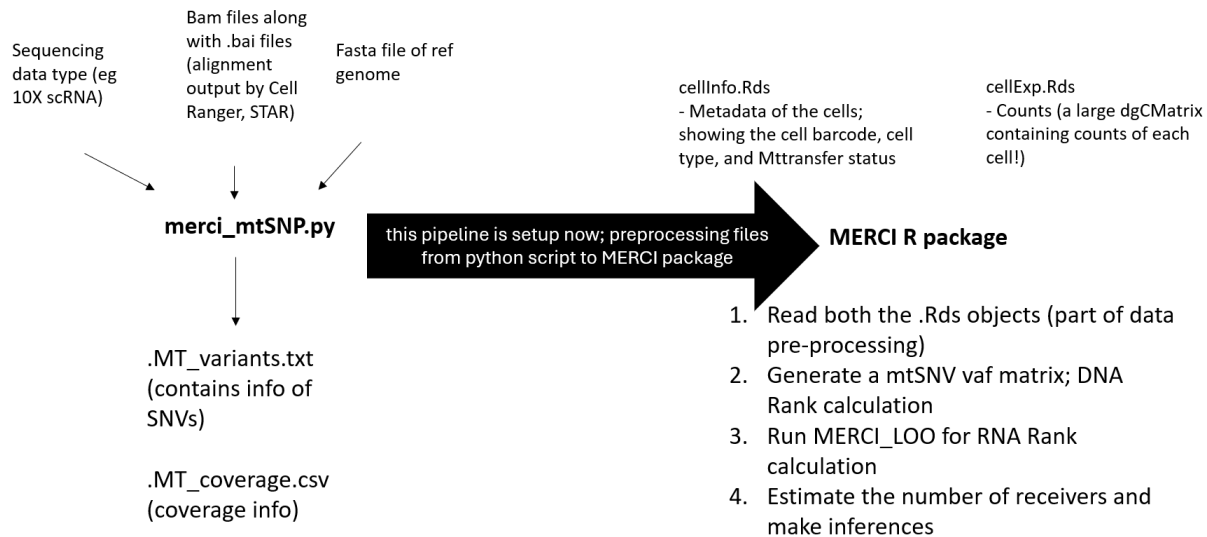


Figure 3. MERCI workflow overview

Next Steps

Currently, we have set up a pipeline to preprocess the files from the mtSNP-module to the MERCI R package. Next, we would like to work on the pipeline upstream of the mtSNP-module. This pipeline would extract .bam files from .sra files. Specifically, we will download the raw sequencing files (.sra) from the NCBI SRA database, convert them to .fastq using the SRA Toolkit, align the reads to the reference genome with an aligner such as HISAT2 or BWA to generate a .sam file, convert the .sam to .bam using samtools, and then sort and index the resulting .bam file for efficient retrieval and visualization.

Next, we have identified multiple datasets where we would like to employ the MERCI technique to identify mitochondrial transfer events. To begin with, we would like to start with datasets which are cocultured, so it would be easy to mark which cells are the receivers and donors. For this we identified co-cultured datasets (GSE181410) (2) and (GSE214569) (3). (GSE181410) is a cocultured colon cancer and cancer-associated fibroblast dataset. Although, this dataset is from Celseq-2 and not 10X, therefore it is highly unlikely we will have good MT-coverage. (GSE214569) is a cocultured adenocarcinoma and macrophage dataset. To use this dataset, we would first need to set up the .sra to bam pipeline, since there is no bam file given by the authors. Once we are able to report and validate the findings of MERCI successfully on these datasets and examine the directionality of mitochondrial transfer within the co-culture system we would like to move to further datasets which include multiple cells. An ideal benchmark would be to set

up a MERCI pipeline for publicly accessible scRNA-seq and scATAC-seq data.

References

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